Noubhar Ahmad

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Education

Clemson University — Clemson, United States
MSc in Mechanical Engineering — *August. 2026 – Expected May. 2028*

University of Engineering and Technology, Lahore (UET) — Lahore, Pakistan
BSc in Mechanical Engineering — *Sept. 2019 – May 2023*
Senior Thesis: Fabrication of 3D Printer & Design Optimization of 3D Printed Parts for Digital Manufacturing.

Experience

Mechanical Engineer — Sept. 2023 – Oct. 2025
Mace Engineering Technologies — *Lahore, Punjab, Pakistan*
- Investigated microstructure (layer adhesion, porosity) and mechanical properties for (FDM), based AM.
- Designed automobile components including rocker covers, cylinder heads, and pistons.
- Performed (Optical Electro Spectrometry) tests to determine chemical composition of H13 and 42CrMo4 steels.
- Conducted computational simulations in ANSYS to analyze thermal properties of steel and aluminum.
- Performed metallographic preparation and microstructure analysis of H13 and 42CrMo4 steels.
- Evaluated experimental and research findings and presented results in weekly manager meetings.

Summer Intern — Summer 2022
Pakistan Aeronautical Complex — *Kamra, Punjab, Pakistan*
- Designed and performed simulation analysis on Twin Ventral Fins for the JF-17 aircraft.
- Investigated material characterization of Ti-6Al-4V alloy.
- Developed prototypes of Twin Ventral Fins using Directed Energy Deposition (DED) additive manufacturing.

Summer Intern — Summer 2022
Heavy Mechanical Complex — *Taxila, Punjab, Pakistan*
- Gained hands-on experience in ultimate tensile testing, bend testing, and compression strength testing of refractory materials.
- Investigated surface finish imperfections in metals during casting processes.

Manufacturing Intern — Summer 2021
Pakistan Railways — *Lahore, Punjab, Pakistan*
- Acquired knowledge of manufacturing processes including CNC machining, casting, forging, and related operations.
- Gained practical experience in the manufacturing of locomotive bodies, fabrication, and welding.

Technical Skills

Design & Simulation: ANSYS, SolidWorks
Laboratory Testing: Tensile, Hardness, (OES), (SEM)
Programming: MATLAB

Research Interests

Manufacturing Process Optimization, Additive Manufacturing, Meta Materials, Intelligent Process control, Material Characterization.

Research Experience

Undergraduate Research Assistant — 2021 – 2022
Department of Mechanical Engineering, UET Lahore — *Lahore, Pakistan*
- Designed intricate components such as gears and shafts using CAD software.
- Investigated the material characterization of polymers, biomaterials, and ceramics.
- Conducted experimental testing including tensile, bending, and torsional testing.
- Investigated stress distribution and fracture potential at gear tooth roots and shaft necks.
- Led a research group of 4 students in conducting literature reviews.

Publications

Published

Tayyab, S., Umair, S., Abu, T., **Noubhar, A.,*** Saboor, Z., & Faris, I. "Fabrication of 3D Printer & Design Optimization of 3D Printed Parts for Digital Manufacturing." *IJESIR Novus*, 2023–2024.

In Preparation

Noubhar, A.,* Shafahat, A. "Stimuli-Responsive 4D-Printed Metamaterials: Materials, Design, Mechanisms, and Multifunctional Applications." *Review*, 2025.

Relevant Courses

Note: *Italicized are non-credit courses.*

Clemson: Advance design methodologies, Design and Manufacturing, Laser based AM, Engineering optimization, Material Selection in Manufacturing Process.

UET: Industrial Materials, Design and Manufacturing, Mechanics of Materials, Finite Element Analysis, Advanced Mechanics of Materials, *Additive Manufacturing*, *Material Characterization*, *3D Printing*

Competitions & Hackathons

Erasmus+ Hackathon on LCA of Sustainable Materials 2025–2026

Silesian University of Technology *Gliwice, Poland*

- Selected a scientific article on LCA of additive manuafacturing, analysed it, and presented findings in a report.
- Completed a course on Life Cycle Assessment and attended a workshop on sustainable materials.

Honors & Awards

2025–2026 – Fully Funded Participant, (HI-EURECA-PRO) Industrial Hackathon, Gliwice, Poland

2022–2023 – Award of Recognition for Services as President, (ASME) UET Lahore

2022–2023 – Award of Recognition for Services as President, (ASME) UET Lahore